

ESG and BRSR Notes (Part 1)

The main problem

MEIL has many projects and companies in India and abroad. The government and SEBI want companies to report on environment, employees and governance. MEIL must collect this data from all projects and make a BRSR report. Collecting so much data by hand is very hard. So they need software that can collect, calculate, check and report this data.

What is MEIL?

MEIL means Mega Engineering and Infrastructure Limited. It is a big infrastructure company. Infrastructure means big physical projects that a country needs, like roads, highways, airports, power plants, water projects, irrigation, oil and gas projects and transport projects.

From the podcast: MEIL has about 250 to 300 live projects in India and about 15 to 20 outside India. It also has completed projects. For some projects it has O&M; contracts.

O&M; means Operations and Maintenance. Example: MEIL builds a power plant. After that, it may have to run the plant for 10 more years, maintain the equipment, and track electricity, water and fuel use.

Why is there a problem?

MEIL is a very big company. Suppose it has 300 projects. Each project must give data like this:

- Diesel: 50,000 litres
- Electricity: 2 lakh kWh
- Water: 5 lakh litres
- Waste: 10 tonnes
- Employees: 200
- Training: 500 hours

That is 300 projects, many data points each, every month or year. This is a huge amount of data. If it is collected by Excel, WhatsApp or email, there will be many problems. So the company needs one central web-based software. Projects enter their data there, and the head office collects, calculates and reports it automatically.

SDG

SDG means Sustainable Development Goals. These are 17 global goals made by the United Nations (UN) to make the world better and sustainable. Examples: reduce poverty, quality education, clean water, clean energy, gender equality, decent work, climate action.

Sustainable means development that helps us today and does not harm future generations. Example: a factory that makes products but also pollutes water without control is not sustainable. A factory that controls pollution, manages waste and saves water is more sustainable.

The UN (United Nations) is an international group of many countries. It sets goals for global issues. SDGs are UN goals.

The 5 Ps of SDG

The 17 SDGs are grouped into 5 Ps. Dr. Ravi mentioned these in the podcast.

- **People:** health, education, equality, less poverty
- **Planet:** pollution, climate, water, waste, natural resources
- **Prosperity:** jobs, economic growth, better living
- **Peace:** peaceful and fair societies
- **Partnership:** countries and organisations working together

From SDG to ESG

The UN made the SDGs. But a company like MEIL cannot just say "we support SDGs". It has to show proof with data that it follows responsible practices. This is where ESG comes in.

ESG

ESG means Environmental, Social and Governance.

- **E (Environmental):** how much the company affects the environment. Examples: carbon emissions, electricity, fuel, water, waste, pollution, renewable energy.
- **S (Social):** what the company does for people. Examples: employees, safety, health, training, diversity, worker welfare, community development.
- **G (Governance):** how properly the company is run. Examples: policies, compliance, ethics, board structure, anti-corruption, risk management, transparency.

Example for a MEIL construction project:

- **E data:** 1 lakh litres diesel used, 5,000 kWh electricity used, 20 tonnes waste, 10 lakh litres water used.
- **S data:** 300 employees, 200 workers got safety training, 5 safety incidents, working hours.
- **G data:** company follows policies, does audits, keeps compliance, handles complaints.

In short: ESG = Environment + People + Company management.

SDG vs ESG

SDG	ESG
Global goals by the UN	A way to check and report a company
17 goals	3 main areas
People, Planet, etc.	Environmental, Social, Governance
Focus on world development	Focus on company responsibility

Easy trick: SDG = the world's goal. ESG = the company's responsible behaviour and data.

BRSR

BRSR means Business Responsibility and Sustainability Report. It is a report where a company tells what it is doing about responsibility and sustainability. It includes environment data, employee data, energy, water, emissions, waste, social information and governance information.

SEBI

SEBI means Securities and Exchange Board of India. It is the authority that controls India's stock market. If a company is listed on the stock market, SEBI has rules about what it must disclose to investors. The podcast talks about SEBI's BRSR guidelines.

Chain to remember: SEBI makes the BRSR rules, companies collect data, and then they make the BRSR report.

A listed company is one whose shares can be bought and sold by the public on a stock exchange. In India the main exchanges are NSE and BSE. Listed companies must make certain disclosures.

MEIL and BRSR

Dr. Ravi said MEIL is currently not legally required to do the BRSR reporting they are discussing. But MEIL is doing it by choice. This is called voluntary. The law has not forced it, but the company wants to do it to show responsible business and to be ready for future rules.

The full picture of the hackathon problem

```
UN
|
17 SDGs
|
Need to measure and report company responsibility
|
ESG
|
BRSR (reporting framework in India)
|
MEIL must prepare BRSR data
|
Hundreds of projects and subsidiaries
|
Huge data, hard to collect by hand
|
HACKATHON PROBLEM:
Build a web-based ESG/BRSR data collection and reporting system
```

What will the software do?

Example: Highway Project A. The project manager logs in. The software asks:

- Electricity used? 50,000 kWh
- Diesel used? 20,000 litres
- Water used? 500,000 litres
- Waste generated? 15 tonnes
- Number of employees? 250

- Number of safety incidents? 3

The manager also uploads proof, like the electricity bill, fuel receipt, water record and waste disposal paper. Then data from all projects goes to the central MEIL system.

Documentary evidence

This means uploading a document as proof with the data. Example: instead of only typing "Diesel = 20,000 litres", the user also uploads the diesel invoice as PDF or image. Then the data can be traced and audited.

```
Diesel consumption: 20,000 litres
Evidence: diesel_invoice.pdf
```

Why the project is data-centric

The core of the project is this flow:

```
COLLECT DATA -> VALIDATE -> CALCULATE -> AGGREGATE -> REPORT
```

Carbon

Carbon emission came up again and again in the podcast. Many activities release greenhouse gases. For reporting, these are shown in one common unit called CO₂ equivalent (CO₂e).

```
Diesel used -> Emission factor -> CO2e
```

Fuel quantity x emission factor = estimated emissions.

Emission factor

An emission factor tells how much greenhouse gas is released by one unit of something. Example: how much gas does 1 litre of diesel release? A scientifically accepted number is used for this.

```
Activity data x Emission Factor = Estimated Emission
```

```
10,000 litres diesel x diesel emission factor = X kg CO2e
```

Important: do not pick a random factor. Take it from a trusted standard or source.

Carbon equivalent

There are many different greenhouse gases. To keep reporting simple, all of them are shown in one common unit, CO₂-equivalent. So the system should not only store "Diesel = 20,000 litres". It should also calculate "Estimated emissions = X kg CO₂e" using the chosen emission factor.

Scope 1, Scope 2, Scope 3

This is a very important part of the project. Dr. Ravi discussed all three scopes.

Scope 1: direct emissions. Emissions from things the company owns or controls. Example: MEIL's construction equipment burns diesel. Trick: Scope 1 = directly me.

Scope 2: emissions from purchased electricity. Example: MEIL buys electricity from the grid. The emissions did not happen at the site, but they happened where the electricity was made. From

the podcast: if the electricity is renewable or green, the emission calculation can be different or lower, depending on the method used.

Scope 3: other indirect emissions in the value chain. Podcast examples: employee travel, train travel, flights, vehicles. Example: an MEIL employee travels by flight. MEIL did not run the plane, but the business travel is linked to the company.

Scope	Simple meaning	Example
Scope 1	Direct emissions	Diesel burnt in company equipment
Scope 2	Indirect emissions from purchased electricity	Grid electricity
Scope 3	Other indirect emissions in the value chain	Employee flight or train travel

Memory trick: 1 = I burn. 2 = I buy electricity. 3 = Others / value chain.

Why Scope 3 gets less focus

Dr. Ravi said that for the hackathon prototype, the main focus should be Scope 1 and Scope 2. Scope 3 can be done at a basic level. Reason: Scope 3 is very broad (flights, trains, taxis, business travel, vehicles, supply chain, purchased goods) and its data and calculation are complicated. So the priority is Scope 1, then Scope 2, then basic Scope 3.

Water

ESG is not only about carbon. Water is also important. The system may need to record:

- Water source: groundwater, river, municipal supply, other
- Quantity: for example 500,000 litres from groundwater

So capture both where the water came from and how much was used.

Waste management

Waste is another big environmental data area. Example: construction waste 10 tonnes, plastic waste 2 tonnes, hazardous waste. The system may need to track:

- Type of waste
- Quantity
- Disposal method
- Recycling or reuse
- Supporting evidence

Fuel

Podcast examples of fuels: petrol, diesel, coal, CNG, LPG. The system must record fuel use.

Example: Fuel type Diesel, quantity 20,000 litres, period August 2026, project Highway Project A. Then emissions can be calculated using the right emission factor.

Electricity

Do not only ask "How much electricity?". Also ask which source it came from (grid, renewable, etc.). Then calculate and report emissions using the right method.

Employee and HR data

This is the Social part. Dr. Ravi said you also have to include HR data. Possible data: number of employees, worker data, training, health, safety, diversity, working conditions, incidents, welfare. So the project is not only environmental. It covers Environmental + Social + Governance.

Governance

Governance means how the company runs properly, ethically and by the rules. Examples: policies, compliance, audits, risk, ethics, anti-corruption, grievance handling, responsible management.

Company structure and subsidiaries

Podcast: MEIL has about 20 to 28 subsidiary companies. A subsidiary is a company controlled by a main (parent) company. Subsidiaries can also have their own companies and projects, so the structure is complicated.

```
MEIL (parent)
|
|-- Subsidiary A
|   |-- Project 1
|   |-- Project 2
|   |-- Project 3
|
|-- Subsidiary B
|   |-- Project 4
|   |-- Project 5
|
|-- Subsidiary C
|   |-- Project 6
|   |-- Project 7
```

Data goes from bottom to top: Project -> Subsidiary -> MEIL -> ESG/BRSR report.

Aggregation

Aggregation means combining data from many projects into one higher-level total. Example: Project A diesel 10,000 L + Project B 20,000 L + Project C 30,000 L = 60,000 litres. The system should do this automatically.

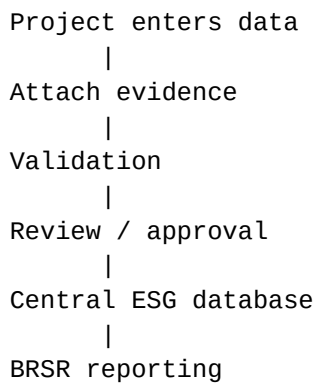
Monthly data

Dr. Ravi said people from about 250 projects should enter data about power and water every month. The project user logs in, picks the project and month (for example September 2026), and enters electricity, fuel, water, waste and so on. Next month, the same again.

Data validation

Validation means checking the entered data. Example: Diesel = -50,000 litres is clearly wrong, so the system should show "Invalid quantity". Also, if someone enters an extremely huge number like 5000000000000000 litres of water, the system can flag it as unusual.

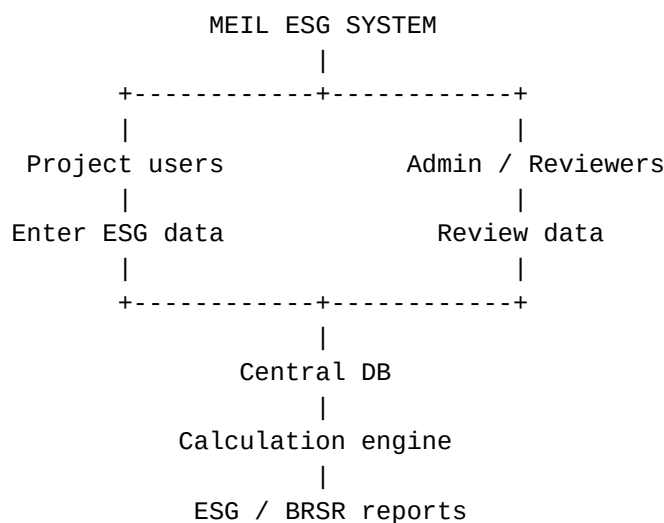
Validation plus documentary evidence gives better data quality. The workflow:



Web-based application

Web-based means it runs in the browser (like Chrome). Projects are in different places, and all of them need to use one central system. So a web application is a good fit.

Basic idea of the system



Do not think about coding yet. First understand the concept.

What the judges want to see

- **Data collection:** projects can enter data.
- **Evidence:** users can upload documents.
- **Calculation:** emissions are calculated from fuel, electricity, etc.
- **Consolidation:** data of hundreds of projects is combined.
- **ESG dashboard:** management gets an overview.
- **Reporting:** structured output for BRSR reporting.

Dr. Ravi's main advice

Advice 1: Understand the problem deeply. Do not just start coding. First understand what ESG is, what BRSR is, what data will be collected, what Scope 1, 2 and 3 are, how the projects are structured, and what the reporting needs are.

Advice 2: Clarify the important deliverables. Be clear on exactly what software you have to deliver.